

1 Solution — GIAM 3.4

Problem 2

1.1 Problem

Find a counterexample to the conjecture that

$$\forall a, b, c \in \mathbb{Z}, \quad a \mid bc \implies a \mid b \vee a \mid c$$

using only powers of 2.

1.2 Answer

$a = 4,$	$b = 2,$	$c = 2.$
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1.3 Verification

All three are powers of 2, as required. Checking the hypothesis and both halves of the conclusion:

claim	check	
$a \mid bc$	$4 \mid 2 \cdot 2 = 4$	TRUE
$a \mid b$	$4 \mid 2$	FALSE
$a \mid c$	$4 \mid 2$	FALSE

The hypothesis holds and the conclusion — a disjunction — fails in both disjuncts. So the universally quantified statement is false, and one instance is all it takes to say so.

1.4 Why One Instance Is Enough

The conjecture is a **universally quantified statement**, so its negation is existential:

$$\neg \left(\forall a, b, c, P(a, b, c) \right) \cong \exists a, b, c, \neg P(a, b, c),$$

and exhibiting one triple that witnesses $\neg P$ is a complete proof of the negation. Note also that $\neg(Q \vee R) \cong \neg Q \wedge \neg R$, which is why both $4 \nmid b$ and $4 \nmid c$ had to be checked — refuting only one disjunct would refute nothing.

1.5 The Picture

Restricting to powers of 2 makes every divisibility a statement about exponents, and the whole situation becomes visible at once.

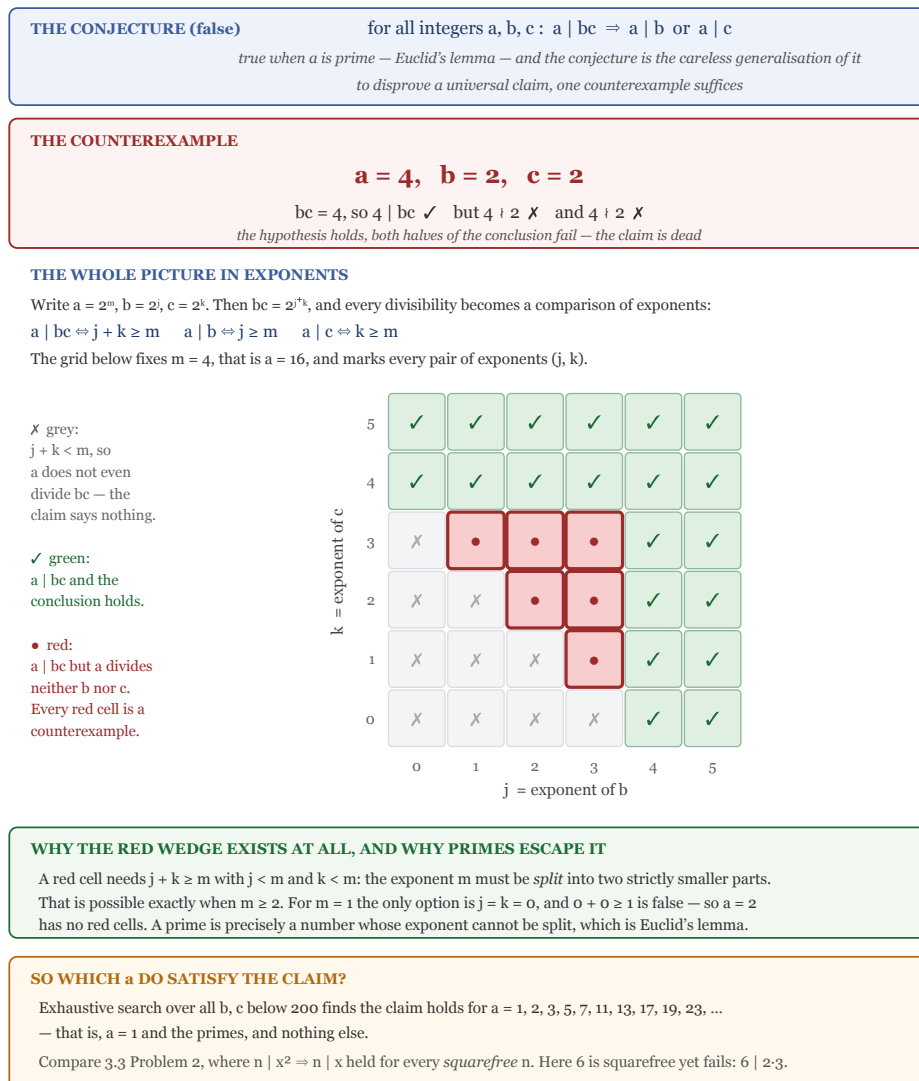


Figure 1.1: A diagram in five bands. The first states the false conjecture: for all integers a, b, c , a divides bc implies a divides b or a divides c ; it is true when a is prime, which is Euclid's lemma, and the conjecture is the careless generalisation of it. To disprove a universal claim, one counterexample suffices. The second band gives the counterexample: a equals 4, b equals 2, c equals 2. Then bc equals 4, so 4 divides bc , but 4 does not divide 2 and 4

does not divide 2. The hypothesis holds, both halves of the conclusion fail, and the claim is dead. The third band explains the exponent picture: write a as 2 to the m , b as 2 to the j , c as 2 to the k ; then bc is 2 to the j plus k , and every divisibility becomes a comparison of exponents — a divides bc if and only if j plus k is at least m , a divides b if and only if j is at least m , and a divides c if and only if k is at least m . A six-by-six grid fixes m equal to 4, that is a equals 16, and marks every pair of exponents j and k from 0 to 5. Cells where j plus k is less than m are grey and carry a cross: a does not even divide bc , so the claim says nothing. Cells where a divides bc and the conclusion holds are green with a tick. Cells where a divides bc but a divides neither b nor c are red with a dot; every red cell is a counterexample, and they form a triangular wedge in the middle of the grid. The fourth band explains why the red wedge exists and why primes escape it: a red cell needs j plus k at least m with j less than m and k less than m , so the exponent m must be split into two strictly smaller parts, which is possible exactly when m is at least 2. For m equal to 1 the only option is j equals k equals 0, and 0 plus 0 at least 1 is false, so a equals 2 has no red cells. A prime is precisely a number whose exponent cannot be split, which is Euclid's lemma. The fifth band asks which a do satisfy the claim: exhaustive search over all b and c below 200 finds it holds for a equal to 1, 2, 3, 5, 7, 11, 13, 17, 19, 23 and so on — that is, a equals 1 and the primes, and nothing else. Compare 3.3 Problem 2, where n divides x squared implies n divides x held for every squarefree n ; here 6 is squarefree yet fails, since 6 divides 2 times 3.

1.6 Remark — Everything Becomes Exponent Arithmetic

The restriction to powers of 2 is not an arbitrary handicap; it is what makes the whole landscape legible. Writing

$$a = 2^m, \quad b = 2^j, \quad c = 2^k,$$

we have $bc = 2^{j+k}$, and since $2^s \mid 2^t$ exactly when $s \leq t$, the three divisibility statements collapse into three inequalities:

$$a \mid bc \iff j + k \geq m, \quad a \mid b \iff j \geq m, \quad a \mid c \iff k \geq m.$$

The conjecture therefore asserts

$$j + k \geq m \implies j \geq m \text{ or } k \geq m,$$

which, stated that baldly, is plainly false — it is the claim that *a sum can only be large if one of its summands is large*. Two medium numbers add to a big one all the time. The smallest instance is $j = k = 1$, $m = 2$, which is exactly $(a, b, c) = (4, 2, 2)$.

1.7 Remark — Why $m \geq 2$ Is Needed, and What That Says About Primes

A counterexample requires

$$j + k \geq m \quad \text{with} \quad j < m \quad \text{and} \quad k < m,$$

that is, m must be *split* into two strictly smaller non-negative parts. That is possible precisely when $m \geq 2$.

For $m = 1$ — the case $a = 2$ — the constraints $j < 1$ and $k < 1$ force $j = k = 0$, and then $j + k = 0 \geq 1$ is false. There is no red cell, confirmed by exhaustive search: $a = 2$ satisfies the conjecture. Likewise $m = 0$, the case $a = 1$, which divides everything.

This is the whole reason primes are special. In the exponent picture, p prime means the exponent to be reached is 1 , and 1 cannot be written as a sum of two smaller non-negative integers other than $0 + 1$ — in which case one of them already equals 1 . **Euclid's lemma is the statement that 1 is unsplitable.** The failure for $a = 4$ is the statement that $2 = 1 + 1$ is not.

1.8 Remark — Exactly Which a Satisfy the Claim

Beyond powers of 2 , one can ask which moduli make the implication true for *all* b, c . Exhaustive search over $0 \leq b, c < 200$ gives

$$a = 1, 2, 3, 5, 7, 11, 13, 17, 19, 23, 29, \dots$$

— precisely 1 together with the primes, and nothing else. Every composite a fails: write $a = uv$ with u and v both strictly between 1 and a , and then $a \mid uv$ while $a \nmid u$ and $a \nmid v$. The counterexample is built directly out of the factorisation.

It is worth putting this beside 3.3 Problem 2, which asked about $n \mid x^2 \Rightarrow n \mid x$:

statement	holds exactly for
$n \mid x^2 \implies n \mid x$	n squarefree
$n \mid bc \implies n \mid b \text{ or } n \mid c$	n prime (or 1)

The second condition is strictly stronger, and the reason is that b and c are *independent*. With x^2 the two factors are forced equal, so the exponent must split evenly, and only a repeated prime factor can cause trouble — hence squarefree suffices. With bc the factors can be chosen separately, so any composite a can be torn apart. The example $n = 6$ separates them: 6 is squarefree, so $6 \mid x^2 \implies 6 \mid x$ holds, yet $6 \nmid 2 \cdot 3$ while 6 divides neither factor.

1.9 Remark — The Conjecture Is a Real Theorem With Its Hypothesis Removed

Statements like this one are worth recognising as *damaged theorems* rather than random falsehoods. The true version is Euclid’s lemma:

*If p is **prime** and $p \mid bc$, then $p \mid b$ or $p \mid c$.*

The conjecture is what is left after deleting the word “prime”. That deletion looks harmless — the letter a still ranges over integers, the statement still parses — but it is doing all the work, and the counterexample locates the damage precisely.

This is a common shape for a false conjecture, and it suggests a useful reflex when confronted with one: rather than hunting blindly, ask *what true statement is this a corruption of?* The hypothesis that was dropped tells you where to look. Here, dropping primality means looking at a composite, and the smallest composite prime power is 4.

The same lemma, incidentally, is what makes unique factorisation work. Rings where Euclid’s lemma fails for irreducible elements are exactly the rings where factorisation is not unique — in $\mathbb{Z}[\sqrt{-5}]$, for instance, 2 is irreducible and divides $6 = (1 + \sqrt{-5})(1 - \sqrt{-5})$ without dividing either factor, and correspondingly $6 = 2 \cdot 3$ has a second factorisation. So the failure exhibited in this exercise, transplanted to a different ring, is not a curiosity but the precise obstruction to unique factorisation.